WAYGROUND formerly Quizizz **Worksheets**

Atomic Structure and Bonding

Total questions: 20

Worksheet time: 10mins

Instructor name: So Korm

Name

Class

Date

1. What sub-atomic particle has a neutral charge?

   a) Electron
   b) Neutron
   c) Nucleus
   d) Proton

2. What type of bonding involves the transfer of electrons?

   a) Ionic
   b) Covalent
   c) Metallic

3. Give the name for atoms of the same element, with the same number of protons but a different number of neutrons.

   a) Ion
   b) Atom
   c) Isotope
   d) Element

4. Where is most of the mass found in the atom?

   a) Nucleus
   b) Empty space
   c) Shells

5. What type of bonding involves the sharing of electrons?

   a) Covalent
   b) Ionic
   c) Metallic

6. Oxygen is in Group 6, what charge will an oxide ion have?

   a) -1
   b) +1
   c) +2
   d) -2

7. Give the formula of the magnesium fluoride.

a) Mg_2F
b) MgF_2
c) 2MgF
d) MgF

8. Na has an atomic number of 11 and a mass number of 23. State the number of protons, neutrons and electrons.

a) P = 11, E = 12, N = 11
b) P = 11, E = 11, N = 12
c) P = 11, E = 11, N = 11
d) P = 11, E = 11, N = 34

9. Give the formula of sodium oxide.

a) NaO
b) Na_2O
c) 2NaO
d) NaO_2

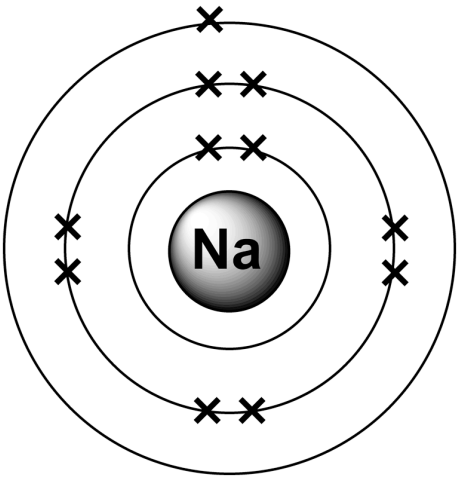
10. State the correct electronic configuration of a Magnesium ion.

a) 2.8.8.2
b) 2.8
c) 2.8.2

11. What type of elements are involved in ionic bonding?

a) Non-metals only
b) Metals and non-metals
c) Metals only

12.



What group is this element in?

a) 3
b) 11
c) 4
d) 1

13. Why does an atom have an overall neutral charge?

a) Equal numbers of electrons and neutrons
b) Equal numbers of protons and neutrons
c) Equal numbers of protons, neutrons and electrons
d) Equal numbers of protons and electrons

14. An atom has an overall positive charge because...

a) there are more protons than neutrons
b) there are more electrons than protons
c) there are more protons than electrons

15. Name the type of bonding in: NH_3, H_2, H_2O, CO_2

a) Ionic
b) Covalent
c) Metallic

16. Why does potassium oxide have the following formula: K_2O?

a) 1 K atom required for every 1 oxygen atom
b) 2 K ions required for every 1 oxide ion
c) 1 K ion required for every 1 oxide ion
d) 2 K atoms required for every 1 oxygen atom

17. State the type of bonding in the following compounds:
NaCl, MgO, LiF, KF

a) Covalent
b) Metallic
c) Ionic

18. Which groups do not take part in ionic bonding?

a) 2 and 0/8
b) 4 and 0/8
c) 5 and 4
d) 3 and 7

19. State the relative mass of an electron

a) 1/2000
b) 1
c) 1/1840
d) 1/1800

20. Carbon is in Group 4, how many covalent bonds can it make?

a) 3

b) 2

c) 4

d) 1

Answer Keys

1. b) Neutron
2. a) Ionic
3. c) Isotope
4. a) Nucleus
5. a) Covalent
6. d) -2
7. b) MgF_2
8. b) P = 11, E = 11, N = 12
9. b) Na_2O
10. b) 2.8
11. b) Metals and non-metals
12. d) 1
13. d) Equal numbers of protons and electrons
14. c) there are more protons than electrons
15. b) Covalent
16. b) 2 K ions required for every 1 oxide ion
17. c) Ionic
18. b) 4 and 0/8
19. c) 1/1840
20. c) 4